Ana Isabel Lopez Murillo

CURRICULUM VITAE

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EDUCATION

UNIVERSITY OF NORTH CAROLINA – CHAPEL HILL — Chapel Hill, North Carolina

Ph.D., Physics and Astronomy — Expected May 2030
Graduate Thesis: A Physics-Anchored AI Model for Predicting Starspot-Induced Variability
Advisor: Dr. Andrew Mann

B.S., Computer Science — May 2025
Undergraduate Thesis: Searching for Transit Timing Variations in Young Transiting Systems
Advisor: Dr. Andrew Mann

AWARDS AND HONORS

NC Space Grant Graduate Research Fellowship *(NC State, NASA)* — 2026
NC Space Grant Graduate Research Fellowship *(NC State, NASA)* — 2025
Robert N. Shelton Award for Outstanding Research *(UNC Chapel Hill)* — 2025
Chancellor's Science Scholar *(UNC Chapel Hill)* — 2021 – 2025
Highly selective merit-based STEM scholarship program
Honors Carolina *(UNC Chapel Hill)* — 2021 – 2025
Competitive, four-year academic program enrolling 10% of each class
Dean's List *(UNC Chapel Hill)* — 2021 – 2025
Renwick Scholar Academic Award *(UNC Chapel Hill)* — 2022, 2023

RESEARCH EXPERIENCE

University of North Carolina – Chapel Hill — Chapel Hill, North Carolina

Graduate Research Assistant, Department of Physics and Astronomy — August 2025 – present
Characterizing young exoplanet masses and atmospheres by developing pipelines for automated analysis procedures of ground-based telescope data.

Undergraduate Research Assistant, Department of Physics and Astronomy — January 2023 – July 2025
Analyzing potential transit time variations (TTVs) of young planetary systems to characterize attributes of exoplanets using Python, statistical analysis, and optimization methods.

RESEARCH INTERESTS

Exoplanet discovery and characterization, software engineering for astronomy applications, exoplanet atmospheres, stellar variability, transit timing variation analysis

PROFESSIONAL EXPERIENCE

Wells Fargo — Charlotte, North Carolina

FC-Ops Intern Analyst — June 2024 – August 2024
Proposed and developed a native Windows application to accelerate ILQC review process by over 10% using WinUI, .NET, and Windows App SDK Frameworks.

TECHNICAL SKILLS

Python, Java, JavaScript, Typescript, RxJS, AngularJS, HTML/CSS, Git, HTTP, Full-Stack Engineering, Docker, SQL, MCMC Analysis, Algorithmic Analysis, Statistical Optimization, BATMAN, Pandas.

PUBLICATIONS

Lopez Murillo, A. I., Mann, A. W., Barber, M.G. *et al*. Searching for Transit Timing Variations in Young Transiting Systems. *AJ* **171,** 63 (2026). https://doi.org/10.3847/1538-3881/ae231a

Barber, M.G., […], Lopez Murillo, A. I. *et al.* TESS Investigation—Demographics of Young Exoplanets (TI-DYE). III. An Inner Super-Earth in TOI 2076. AJ **170**, 32 (2025). https://doi.org/10.3847/1538-3881/add7db

Barber, M.G., […], Lopez Murillo, A. I. *et al.* A giant planet transiting a 3-Myr protostar with a misaligned disk. *Nature* **635**, 574–577 (2024). https://doi.org/10.1038/s41586-024-08123-3

Thao, P.C., […], Lopez Murillo, A.I. *et al*. TESS Hunt for Young and Maturing Exoplanets (THYME). X. A Two-planet System in the 210 Myr MELANGE-5 Association. *AJ* **168,** 41 (2024). https://doi.org/10.48550/arXiv.2406.05234

PRESENTATIONS

Talks

North Carolina Space Grant Research Symposium *(Raleigh, NC)* — Spring 2026
Chancellor's Science Scholars Research Symposium *(Chapel Hill, NC)* — Spring 2025

Posters

Cool Stars 23 Conference *(Tokyo, Japan)* — Summer 2026
Chancellor's Science Scholars Research Symposium *(Chapel Hill, NC)* — Spring 2024, Fall 2023

LEADERSHIP AND SERVICE

Director of Design, *Pearl Hacks*, Marketing Team — 2023-2025
Tutor, *University of North Carolina at Chapel Hill*, Math Department — 2022-2024
Wiki Lead, *UNC iGEM*, University of North Carolina at Chapel Hill — 2023-2024
Teaching Assistant, *ISLA Los Sábados Program*, Weekend Program — 2022